

Brownian Motion / Wiener Process

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First, an example

Before the abstract definition, build W_t by hand. Fix a horizon $T = 1$ and a step $\Delta t = 1/N$ for some large N . Draw iid standard Gaussians $Z_1, \dots, Z_N \sim \mathcal{N}(0, 1)$ and define $W_0 = 0$ and recursively

$$W_{(k+1)\Delta t} = W_{k\Delta t} + \sqrt{\Delta t} Z_k.$$

Read aloud: W at time k plus one delta t equals W at time k delta t plus square root of delta t times Z sub k . By the sum-of-Gaussians rule, at any fixed grid time $t = k\Delta t$, $W_t \sim \mathcal{N}(0, t)$, so $\text{Var}(W_t) = t$ and $\text{Cov}(W_s, W_t) = \min(s, t)$. Letting $N \rightarrow \infty$ (with linear interpolation between grid points) gives a continuous random function on $[0, 1]$ — the sample path of Brownian motion. These paths are continuous but *nowhere differentiable*: each $\sqrt{\Delta t}$ increment is huge compared to Δt , so the path wiggles violently on every scale.

1 Definitions

Definition 1 (Standard Brownian motion / Wiener process). A stochastic process $\{W_t\}_{t \geq 0}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}$ is called a *standard Brownian motion* (or *Wiener process*) if:

1. $W_0 = 0$ almost surely.
2. For all $0 \leq s < t$, the increment $W_t - W_s$ is Gaussian: $W_t - W_s \sim \mathcal{N}(0, t - s)$.
3. **Independent increments:** for any $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are mutually independent.
4. The sample path $t \mapsto W_t(\omega)$ is continuous in t for $\mathbb{P}$ -almost every $\omega \in \Omega$.

Remark 2. Multidimensional Brownian motion $W_t \in \mathbb{R}^d$ consists of d independent copies $W_t = (W_t^{(1)}, \dots, W_t^{(d)})$, each a standard real Brownian motion.

Definition 3 (Natural filtration). The *natural filtration* of $\{W_t\}$ is the increasing family of σ -algebras $\mathcal{F}_t^W := \sigma(W_s : 0 \leq s \leq t)$. We say a process is adapted to $\{\mathcal{F}_t^W\}$ if it is measurable with respect to $\mathcal{F}_t^W$ at each time t .

2 The covariance function

The defining property of Brownian motion that powers nearly every computation in stochastic calculus is its covariance kernel:

$$\mathbb{E}[W_s W_t] = \min(s, t).$$

Read aloud: expectation of W sub s times W sub t equals minimum of s and t . This makes $\{W_t\}$ a centered Gaussian process; in fact this covariance *together with* Gaussianity and the right continuity modification characterizes Brownian motion uniquely in distribution.

Theorem 4 (Covariance of Brownian motion). *Let $\{W_t\}_{t \geq 0}$ be a standard Brownian motion. Then for all $s, t \geq 0$,*

$$\mathbb{E}[W_s W_t] = \min(s, t).$$

Read aloud: expectation of W sub s times W sub t equals minimum of s and t .

Proof. By symmetry of the claim, we may assume without loss of generality that $0 \leq s \leq t$, so that $\min(s, t) = s$.

Step 1: Decompose using independent increments. Write

$$W_t = W_s + (W_t - W_s),$$

Read aloud: W sub t equals W sub s plus the increment W sub t minus W sub s . which is valid by definition. Substituting into the product gives

$$W_s W_t = W_s (W_s + (W_t - W_s)) = W_s^2 + W_s (W_t - W_s).$$

Read aloud: W sub s times W sub t equals W sub s squared plus W sub s times the increment W sub t minus W sub s . Taking expectations and using linearity,

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s (W_t - W_s)]. \quad (1)$$

Read aloud: expectation of W sub s times W sub t equals expectation of W sub s squared plus expectation of W sub s times the increment.

Step 2: Evaluate the first term. By Definition 1 (applied with the increment $W_s - W_0 = W_s$, since $W_0 = 0$ a.s.), we have $W_s \sim \mathcal{N}(0, s)$. Hence

$$\mathbb{E}[W_s^2] = \text{Var}(W_s) + (\mathbb{E}[W_s])^2 = s + 0 = s.$$

Read aloud: expectation of W sub s squared equals variance of W sub s plus the square of its mean, which is s plus zero equals s .

Step 3: The cross term vanishes by independence. Again by Definition 1, the increments $W_s - W_0 = W_s$ and $W_t - W_s$ are independent (these correspond to disjoint time intervals $[0, s]$ and $[s, t]$). Independence implies the expectation of the product is the product of expectations:

$$\mathbb{E}[W_s (W_t - W_s)] = \mathbb{E}[W_s] \cdot \mathbb{E}[W_t - W_s] = 0 \cdot 0 = 0,$$

Read aloud: expectation of W sub s times the increment equals expectation of W sub s times expectation of the increment, which is zero times zero equals zero. since both factors are centered Gaussians with mean 0.

Step 4: Combine. Plugging Steps 2 and 3 back into (1),

$$\mathbb{E}[W_s W_t] = s + 0 = s = \min(s, t),$$

Read aloud: expectation of W sub s times W sub t equals s plus zero equals s equals minimum of s and t . which proves the theorem for $s \leq t$. The case $t \leq s$ follows by symmetry of the inner product, completing the proof. $\square$

Remark 5 (Why this matters). Theorem 4 immediately yields the variance of any linear functional of W . For a deterministic function $f \in L^2([0, T])$, the Wiener integral $\int_0^T f(s) dW_s$ is Gaussian with mean 0 and variance $\int_0^T f(s)^2 ds$ — a direct consequence of the kernel $\min(s, t)$ and the Itô isometry. This is the foundation on which the entire theory of stochastic integration is built.

3 Further properties (stated)

Proposition 6 (Scaling / self-similarity). *For any $c > 0$, the rescaled process $\{c^{-1/2}W_{ct}\}_{t \geq 0}$ is also a standard Brownian motion. Equivalently, $\{W_{ct}\} \stackrel{d}{=} \{\sqrt{c}W_t\}$ as processes.*

Proposition 7 (Martingale property). *The process $\{W_t\}$ is a martingale with respect to its natural filtration: $\mathbb{E}[W_t \mid \mathcal{F}_s^W] = W_s$ for $s \leq t$. Moreover, both $W_t^2 - t$ and $\exp(\lambda W_t - \frac{1}{2}\lambda^2 t)$ ($\lambda \in \mathbb{R}$) are also martingales.*

Proposition 8 (Path regularity). *Almost surely, the sample path $t \mapsto W_t$ is continuous but nowhere differentiable, and has unbounded total variation on every nontrivial interval. Its quadratic variation on $[0, t]$ equals t .*